THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An isosceles triangle has equal sides 30 cm and base 36 cm. Its height (in cm) is: 1

(a) 24 **(b)** 23 **(c)** 12 **(d)** 25

2. A rectangle has a diagonal of 30 cm and one side 18 cm. Its perimeter (in cm) is: 1

(a) 96 **(b)** 86 **(c)** 42 **(d)** 84

3. The side of a rhombus is 30 cm and one diagonal is 36 cm. The other diagonal (in cm) is: 1

(a) 24 **(b)** 48 **(c)** 50 **(d)** 49

4. Two vertical poles of heights 40 m and 22 m stand 24 m apart on level ground. The distance between their tops (in m) is: 1

(a) 28 **(b)** 31 **(c)** 42 **(d)** 30

5. The area of a right triangle with hypotenuse 30 cm and one leg 18 cm (in cm^2) is: 1

(a) 234 **(b)** 216 **(c)** 432 **(d)** 270

6. Which set of numbers is a Pythagorean triplet? 1

(a) 33, 44, 55 **(b)** 33, 46, 55 **(c)** 34, 44, 55 **(d)** 33, 44, 56

7. An isosceles triangle has equal sides 39 cm and base 30 cm. Its height (in cm) is: 1

(a) 37 **(b)** 35 **(c)** 36 **(d)** 24

8. A rectangle has a diagonal of 39 cm and one side 15 cm. Its perimeter (in cm) is: 1

(a) 51 **(b)** 102 **(c)** 108 **(d)** 104

9. The side of a rhombus is 39 cm and one diagonal is 30 cm. The other diagonal (in cm) is: 1

(a) 36 **(b)** 48 **(c)** 74 **(d)** 72

10. Two vertical poles of heights 35 m and 20 m stand 36 m apart on level ground. The distance between their tops (in m) is: 1

(a) 37 **(b)** 51 **(c)** 40 **(d)** 39

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. An isosceles triangle has equal sides of 30 cm and base 36 cm. Find its height. 2

12. The diagonal of a rectangle is 40 cm and one side is 24 cm. Find the other side and the area. 2

13. The side of a rhombus is 50 cm and one diagonal is 60 cm. Find the other diagonal. 2

14. An isosceles triangle has equal sides of 39 cm and base 30 cm. Find its height. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 25 m long reaches a point 24 m up a wall. The foot is then pulled away so that it is 15 m from the wall. How far down does the top of the ladder slip? 3

16. Show that the triangle with sides 18 cm, 24 cm and 30 cm is right-angled and find its area. 3

17. The altitude drawn to the hypotenuse of a right triangle with legs 18 cm and 24 cm: find the hypotenuse and the length of this altitude. 3

OR

A ladder 25 m long reaches a point 20 m up a wall. The foot is then pulled away so that it is 20 m from the wall. How far down does the top of the ladder slip? 3

18. Show that the triangle with sides 15 cm, 36 cm and 39 cm is right-angled and find its area. 3

OR

The altitude drawn to the hypotenuse of a right triangle with legs 15 cm and 36 cm: find the hypotenuse and the length of this altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A man walks 18 m east and then 24 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 40 cm and base 48 cm. Find its height, area and perimeter. 5

20. A man walks 15 m east and then 36 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 35 cm and base 42 cm. Find its height, area and perimeter. 5

END OF QUESTION PAPER — Total: 40 Marks

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